

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0957

Hudson 138 kV
March 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0957. The Interconnection Customer proposes a 421.0 MW natural gas combined cycle facility interconnecting at the Hudson 138 kV substation in the New Jersey area. The request was received on March 04, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 263 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,425,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
5075 HOSP. DAMAS – ADAMS ST 138.0 kV L	15TH STREET – WHITE ROCK-NIKE 138.0 kV	150.7	286	37.3
ADAMS ST – HUDSON 138.0 kV	WHITE ROCK-NIKE – 15TH ST 138.0 kV Line	145.6	311	37.3
LYONS – 5075 HOSP. DAMAS 138.0 kV	WHITE ROCK-NIKE – 15TH STREET 138.0 kV	144.9	274	37.3
IRVINGTON – LYONS 138.0 kV	WHITE ROCK-NIKE – 15TH STREET 138.0 kV Line	129.8	257	37.3
COOK RD SUBSTATION – CLAY STREET 138.0 kV L	MULTITRADE OF PITTSYLVANIA LP – SNE - SANSUM N ELEC 138.0 kV	128.8	245	17.9
15TH STREET – CLAY STREET 138.0 kV L	ADAMS ST – HUDSON 138.0 kV	125.7	463	48.3
BELLEVILLE SUBSTATION – COK RD 138.0 kV	MULTITRADE OF PITTSYLVANIA LP – SNE - SANSUM N ELEC 138.0 kV Line	123.3	234	17.9
WHITE ROCK-NIKE – 15TH ST 138.0 kV Line	ADAMS ST – HUDSON 138.0 kV	122.7	502	48.3
JEFFERSONVILLE KENTUCKY AVENUE – IRVINGTON 138.0 kV	WHITE ROCK-NIKE – 15TH STREET 138.0 kV	118.3	261	37.3
ATESIAN – BELLEVILLE 138.0 kV	MULTITRADE OF PITTSYLVANIA LP SUBSTATION – SNE - SANSUMN ELEC 138.0 kV	116.2	206	17.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
5075 HOSP. DAMAS – ADAMS ST 138.0 kV	Rebuild line with higher-capacity conductor	\$1,767,000	55% (\$979,000)
ADAMS ST – HUDON 138.0 kV Line	Rebuild line with higher-capacity conductor	\$825,000	55% (\$457,000)
LYONS – 5075 HOSP. DAMAS 138.0 kV	Rebuild line with higher-capacity conductor	\$5,955,000	55% (\$3,299,000)
IRVINGTON – LYONS 138.0 kV	Rebuild line with higher-capacity conductor	\$1,651,000	55% (\$914,000)
COOK RD – CLAY ST 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,444,000	46% (\$2,040,000)
CLAY STREET – 15TH ST 138.0 kV	Rebuild line with higher-capacity conductor	\$1,151,000	51% (\$583,000)
BELLEVILLE SUBSTATION – COK RD 138.0 kV	Rebuild line with higher-capacity conductor	\$1,081,000	46% (\$496,000)
WHITE ROCK-NIKE – 15TH ST 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,924,000	51% (\$975,000)
JEFFERSONVILLE KENTUCKY AVENUE – IRVINGTON 138.0 kV	Rebuild line with higher-capacity conductor	\$1,411,000	55% (\$781,000)
ARTESIAN – BELLEVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,963,000	46% (\$901,000)
	Total allocated to project		\$11,425,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.